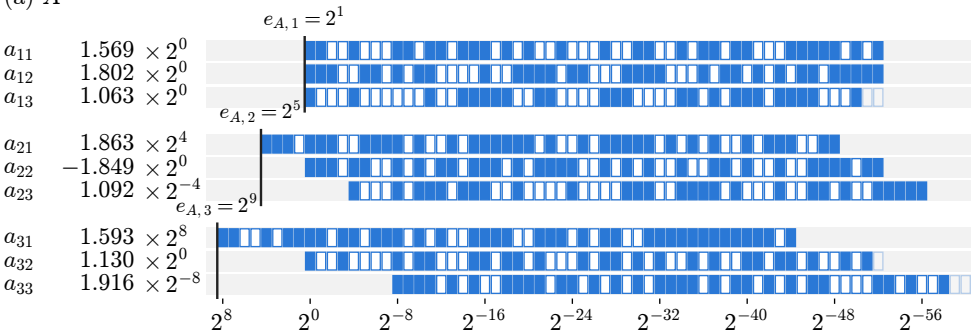
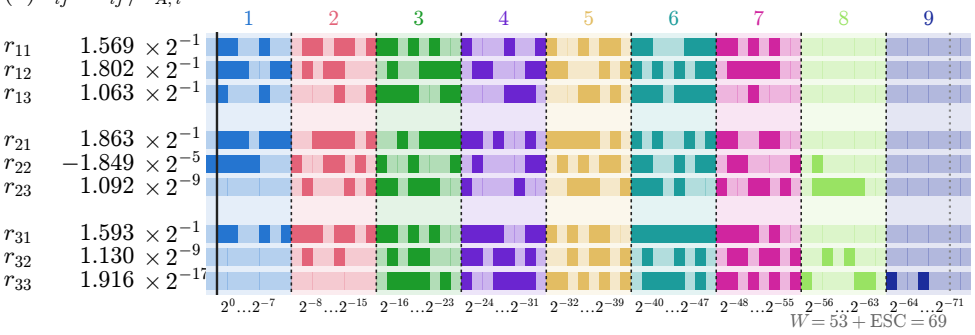
(a) A



(b) $r_{ij} = a_{ij} \,/\, e_{A,i}$



(c) INT8 slices, all stored signed: shared exponent per row, then integers

$$
A^{(1)} \begin{bmatrix} 2^{-6} & 100 & 115 & 68 \\ 2^{-2} & 119 & -7 & 0 \\ 2^{2} & 102 & 0 & 0 \end{bmatrix}
+ A^{(2)} \begin{bmatrix} 2^{-14} & 110 & 88 & 10 \\ 2^{-10} & 61 & -101 & 70 \\ 2^{-6} & -18 & 72 & 0 \end{bmatrix}
+ A^{(3)} \begin{bmatrix} 2^{-22} & -43 & 79 & -13 \\ 2^{-18} & 48 & -111 & -35 \\ 2^{-14} & -43 & 89 & 123 \end{bmatrix}
+ A^{(4)} \begin{bmatrix} 2^{-30} & -118 & 100 & 14 \\ 2^{-26} & -45 & 67 & -124 \\ 2^{-22} & -12 & -38 & -97 \end{bmatrix}
$$

$$
+ A^{(5)} \begin{bmatrix} 2^{-38} & -105 & -58 & 27 \\ 2^{-34} & -5 & 87 & 58 \\ 2^{-30} & -83 & 45 & -43 \end{bmatrix}
+ A^{(6)} \begin{bmatrix} 2^{-46} & -56 & -85 & -17 \\ 2^{-42} & -90 & -53 & -23 \\ 2^{-38} & 0 & 74 & 123 \end{bmatrix}
+ A^{(7)} \begin{bmatrix} 2^{-54} & -44 & 124 & 16 \\ 2^{-50} & -52 & 97 & -102 \\ 2^{-46} & -12 & 109 & -42 \end{bmatrix}
+ A^{(8)} \begin{bmatrix} 2^{-62} & 0 & 0 & 0 \\ 2^{-58} & 0 & 64 & 124 \\ 2^{-54} & 0 & 40 & -121 \end{bmatrix}
$$

$$
+ A^{(9)} \begin{bmatrix} 2^{-70} & 0 & 0 & 0 \\ 2^{-66} & 0 & 0 & 0 \\ 2^{-62} & 0 & 0 & -112 \end{bmatrix}
$$